

RoboCon Oxfordshire 2027
Quidditch

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Clarifications

To ask any questions or for clarifications, please feel free to email support@roboconoxon.org.uk. In all cases, the rules on the website are updated and reflect the most recent state of the game and supersede previous versions including the hard copy. Should we feel any clarifications to the rules need to be made, they will be published to the website and we will inform all teams by email. The rules used during the competition will be the rules published on the official docs website: docs.roboconoxon.org.uk and these rules supersede any other format including but not limited to printed copies of the rules.

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1. Awards

1.1 - The Snitchers

This prize is awarded to teams ranking the first, second and third place in the competition according to their league points.

1.2 - Bimbus 2000

This prize is awarded to the team with the most technically advanced robot, regardless of the mechanisms working or not, trying to implement such mechanisms would be counted into consideration.

1.3 - TeamFlarePrize

This prize is awarded to the team with the most team and robot flare. Things like the overall appearance of the robot will be taken into account.
Tip: Matching cloth with the theme of your robot would help.

1.4 - DifficultyPrize

This prize is awarded to the team who demonstrates the most perseverance in the face of adversity.

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2. Specification

2.1 Arena Specification

[Insert Arena Diagram Here!]

Figure 1 - Top-down diagram of the arena

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2.2 Game Specification

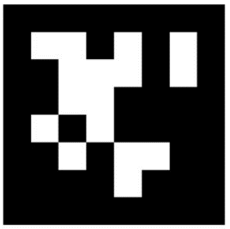
2.2.1 Hoops
2.2.2 Bludgeons
2.2.3 Balls
2.2.4 The Snitch

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2.3 Apriltags Markers

2.3.1. The arena walls and Supplies involved in the game are labelled with AprilTag markers. Each marker pattern encodes a number.

2.3.2. Each marker number is associated with a specific feature of the arena and has an associated size. An example AprilTag can be seen below



2.3.3. Sample markers can be printed on a black and white printer, and their designs can be downloaded from RoboCon Oxfordshire's documentation website, https://docs.roboconoxon.org.uk.

2.3.4. The markers on the arena wall will be 200 x 200 mm in size. They will be 50 mm off the floor.

2.3.5. The markers on the Supplies will be 100 x 100 mm. Supply crates are identified by AprilTags 0 to 23 and Supply drops are identified by AprilTags 24 to 31. Example AprilTags are included in the kit.

2.3.6. Each item in the arena is identified by its own unique number.

2.3.7. The Supplies, arena objects and their corresponding identification numbers are as follows:

Feature	April tag Number range (inclusive)	Colour
Supply crates	0-23 (Of which four are in the arena)	Orange
Supply drops	24-31 (Of which four will be dropped into arena)	Purple
Arena Walls	100 - 102, 121 - 123	Red
	103 - 108	Yellow
	109 - 114	Green
	115- 120	Blue

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3. Rules and Regulations

3.1. The refereename has the final say in any circumstance.
3.2. Robots will be referred to as robotname.
3.3. No remote-controlled systems are permitted.
3.4. RoboCon Oxfordshire is a non-contact competition, and no robotname should be engineered with harming other robotname in mind.
3.5. robotname may not deliberately damage anything. This includes the arena, supply crates, and other robotname. At the refereename's discretion, teams who engage in collisions deliberately or do not take sufficient precautions to avoid collisions may be disqualified from rounds until the issue has been resolved. The refereename decides whether or not this action was deliberate.
3.6. robotname may not deliberately leave debris in the arena.
3.7. Teams must not imitate any features of the arena in a way that could potentially confuse other teams' robotname.
3.8. RoboCon Oxfordshire reserves the right to examine your robotname's software and hardware at any time, to verify that rules are being followed.
3.9. Assistance from RoboCon Oxfordshire is provided with no guarantees.
3.10. All kits handed out by RoboCon Oxfordshire remain the property of RoboCon Oxfordshire. The kit must be returned to RoboCon Oxfordshire at the end of the competition in acceptable condition.
3.11. robotname must pass a safety inspection by a RoboCon Staff before they are permitted in the arena.
3.12. At the beginning of each round, robotname must fit into a cube with internal dimensions of 400 x 400 mm on each side. During the round, the robotname may extend beyond this size.
3.13. For everyone's safety, the robotname's power switch must always be red and easily accessible while the robotname is active – including throughout the game.
3.14. The robotname is to be started in the arena using a physical start button. You must position this where a team member can reach it from outside of the arena.
3.15. You may use custom hardware to enhance your robotname's electronics, but all power must be drawn from the connectors on the Brain.
3.16. All wires connected to the robotname's ground (0V line) must be black. Black wires must not be used for anything else. It is strongly recommended that all wiring is neat and removable, as this will reduce the time required to debug problems, and teams may be asked to tidy their wiring before a member of RoboCon Oxfordshire will approach any issues with their robotname.

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3.17. All electronics should be securely fixed to the robotname (not able to be shaken loose) and should also be easily removable if required.
3.18. It must not be possible to injure oneself on the robotname. This may be tested using a Frankfurter sausage to simulate a finger – anything that could cause harm, such as high-speed rotating parts, should be suitably shielded.
3.19. The lithium polymer battery provided in the kit must be shielded from mechanical and thermal harm. This includes ensuring that it is protected from harm in the case of accidental collision with another Tribute. Teams found to be in violation of this rule will have their batteries confiscated until they have demonstrably rectified the issue.
3.20. robotname may not include additional radio transmitters or receivers to those in the Brain.
3.21. Attaching a GoPro or similar small video recorder to your robotname to record a round is permitted, but it must be powered by its own internal batteries and cannot be connected to any of the other electronics of the robotname. It must fit within the internal dimensions defined in Section 3.1.
3.22. A USB port must remain free and accessible for use during rounds. If you want to use a USB device such as an Arduino, you will need to provide your own USB hub.
3.23. The Brain is supplied as a sealed unit. It must not be tampered with. If you need access to the contents of the Brain, you must contact us so that we can help you with your plans.
3.24. If teams wish to use batteries, chargers, or cables other than the ones provided with the kit, they must seek approval from RoboCon Oxfordshire through support@roboconoxon.org.uk first.

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4. Game Round Specifications

4.1. A round of the game lasts for 180 seconds
4.2. There will be no more than 4 robotname in each round.
4.3. A round may be ended prematurely if all participating teams state to the refereename that they are happy for the round to end.
4.4. Scoring will begin 10 seconds after the end of each round.
4.5. There must be no teams in the arena during the minute before the round starts. robotname must be placed in the arena before this time.
4.6. Teams must not, under any circumstances, enter any body part or object into the arena during a round, except to press the start button on their robotname at the beginning of each round.
4.7. Interfering with any robotname, boxname, or parts of the arena during the round will result in the offending team's disqualification, and they will be asked to leave the arena.
4.8. At the end of a round, the refereename will total the score for each competing team. Teams must not enter the arena or touch any robotname or game items in the arena during this time in order to ensure fair scoring. Touching any robotname or item before the refereename 'releases' the arena may result in the offending team's league points for the round being forfeit.
4.9. Scoring is judged according to the state of the arena at the end of the round.
4.10. robotname must not be programmed to interfere with other robotname. Accidental contact between robotname is expected – we reserve the right to review your code to ensure it complies with this rule.
4.11. A robotname is in possession of a boxname if it touches or encloses the boxname. If a robotname possessing a boxname is enclosed by another robotname, then the innermost robotname is considered in possession of the boxname and the outer robotname(s) is not.
4.12. Scoring is as follows (note that conditions not specified with '**each**' refer to points conceded only once to each team upon meeting the condition):

Condition	Score	
Tribute leaves the Home Area	+1	
Tribute has possession of at least one Supply	+1	
Each Supply inside your Sector	+1 for crate	+2 for drop
Each Supply inside your Sector's Home Area	**+**2 for crate	+3 for drop

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Each Supply in a stack of 2 or more	**Scores relating to stacked crates** × 4

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5. League Points and Knockout Rounds

5.1 League Points

5.1.1. Only teams with robotname present in the area at the start of a round can score league points from that round.

5.1.2. League point scoring:

Score	League points
Highest score	8 League points
Second highest score	6 League points
Third highest score	4 League points
Fourth highest score	2 League points

5.1.3. In the case of a draw, in which two or more teams score the same, each of the teams will gain the average number of points of their places. For instance, if two teams score equally and are therefore in joint first place, they will both score the mean of the first and second place scores – 7 league points.

5.1.4. Teams whose robotname were disqualified from a round will earn no league points for the round.

5.2 Knockout Rounds

5.2.1. Once the league has been completed, a knockout competition will begin. The positions of the teams in the league will seed the positions of teams in the knockout rounds.

5.2.2. Each round in the knockout competition involves up to 4 teams. The teams that come 1st and 2nd in each knockout round will continue to the next round of the knockout.

5.2.3. In case of a tie in a knockout round, the team ranked highest in the league will go through.

5.2.4. If there is a tie in the final for first place, then a rematch may be played at the refereename's discretion.

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6. Kit Return

Each team will be provided with a kit which contains a disclaimer form detailing your obligations with respect to assembly, use and return of the kit, which is lent to you for the duration of RoboCon Oxfordshire 2027. Each team is issued with a kit with the following parts, which is to be returned at the end of the competition.

- **Electronic Kit:**
  - ☐ 1x RoboCon Brain
  - ☐ 1x Power switch- latching (red switch, black connector)
  - ☐ 1x Start Button- momentary (Black switch, green connector)
  - ☐ 1x 6-pin GPIO connector
  - ☐ 1x 2-pin 12V Accessory connector
  - ☐ 2 x 2-pin motor connector
- **Batteries**:
  - ☐ 2 x 3S (11.1V nominal) LiPo Battery
  - ☐ 1 x B3 PRO imaxRC Compact LiPo Charger
- **Computer**:
  - ☐ Linux laptop - checked and updated
  - ☐ Power brick and lead
- **Other**:
  - ☐ 1x 12L Really Useful Box
  - ☐ 1x Really Useful Box Lid
  - ☐ 1x screwdriver set
- **Components**:
  - ☐ 1x components box
  - ☐ 1x IR sensor
  - ☐ 1x RGB LED
  - ☐ 1x LED
  - ☐ 1x contact switch
  - ☐ 2x male to male connectors
  - ☐ 4x male to female connectors
  - ☐ 1x ultrasonic sensor
  - ☐ 1x pink micro servo 9g MG90S

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- **Additional items to be returned if still in working order:**
    - ☐ 1x robot perspex kit
    - ☐ 1x cardboard box with AprilTag
    - ☐ 1x retroflective tape sample
    - ☐ 2x motors with connectors
    - ☐ 6 x AprilTags

If possible, the kit should be returned at the competition, but in no case should it be returned later than 14 days after the competition. If you wish to keep the kit beyond that, this must be arranged prior to the competition date via an email to support@roboconoxon.org.uk

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Thanks

We would like to thank Dr. Peter Willis, Mr. Oliver Lomax, Mr. Graeme May, all the student volunteers and the partners listed below, without whom we would not have been able to run this competition.

















“When students design, build, program and test their own Robot, and then repeat this cycle, solving the problems that have emerged and designing improvements, they are taking part in Engineering. It is a process that involves teamwork skills and project management, as well as combining technical expertise from Design and Technology, Electronics, and Computer Science.”,

~Dr. Peter Willis, Head of Computer Science at Abingdon School





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